# Python Programming (CST-005) - Unit 4

Unit 4 - Functions, Modules & Packages - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

**What this unit covers:** writing your own functions, organising code into modules and packages.

**Key ideas:** Define with **def name(params):** and send results back with **return**. Arguments: positional, keyword (name=value), **default** values, and variable-length ***args** (tuple) and ****kwargs** (dict). Python arguments are passed by object reference - mutable defaults bite, so remember the "don't use [] as a default" rule. **Scope:** local vs global (global keyword). **lambda** makes a one-line anonymous function; map/filter/reduce apply functions over sequences. Recursion: a function calling itself with a base case - factorial and Fibonacci are the standard examples. A **module** is a .py file you import; a **package** is a folder of modules. import forms: import math, from math import sqrt, import numpy as np.

**Exam pointers:** write-a-function questions (factorial by recursion, prime check, palindrome) and *args/**kwargs definitions are standard. Theory: actual vs formal parameters, local vs global with an example, and module vs package in one line each.

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Made by Anas. Free forever. Watch the linked videos on the site after reading.